Interactive Systems

Los Del DGIIM, losdeldgiim.github.io

Doble Grado en Ingeniería Informática y Matemáticas
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Índice general

1. Research Methods for Interactive Systems Design

When designing interactive systems, it is crucial to understand the design space:

- Users: needs and requirements
- Situation: what is the current state of the problem
- Context: location, legislation, culture, etc.

There are different types of research methods:

- Quantitative / Qualitative: Quantitative methods involve collecting and analyzing numerical data, while qualitative methods focus on understanding human behavior and the reasons behind it.
- In situ / In Laboratory: In situ research is conducted in the natural environment of the users, while laboratory research takes place in a controlled setting.
- Unstructured / Structured: Unstructured research allows for more flexibility and exploration, while structured research follows a predefined plan and methodology.

In order to design effective interactive systems, *triangulation* is often used, which involves combining multiple research methods to gain a comprehensive understanding of the design space. During the following sections, we will explore various research methods that can be employed to gather all the needed information.

1.1. Ethnography

Ethnography is a research method that is founded on participation rather than observation, with the goal of not only understanding what people do but also why they do it and how they feel about it. It involves immersing oneself in the users' environment to gain insights into their behaviors, motivations, and experiences.

- It is a qualitative, in situ and unstructured research method that uses induction.

There are two main types of ethnography:

Observing Participation The researcher observes the users' activities and interactions without actively engaging in them. This allows for an objective perspective on the users' behaviors and experiences.

Participating Observation The researcher is actively involved in the users' activities and interactions, allowing for a deeper understanding of their experiences and perspectives. This method has the risk of going "native", where the researcher becomes too immersed in the users' environment and loses objectivity.

Some methods used in ethnography include:

- Field Diary: Personal notes taken by the researcher during their immersion in the field, capturing observations, thoughts, and reflections.
- Cameras: Used to document the environment and interactions, providing visual data for analysis. One important disadvantage is the *Hawthorne Effect*, which refers to the phenomenon where individuals modify their behavior in response to being observed, potentially leading to biased data.
- Diary Studies: Participants keep a diary of their activities and experiences.
- People receive package with material for documentation: Participants are given materials to document their experiences, such as cameras or notebooks. After a certain period, the researcher collects the materials and analyzes the data to gain insights into the users' behaviors and experiences.

1.2. Action Research

AR is a research method that is based on the idea of taking action to solve a problem while simultaneously conducting research to understand the problem and its context. As opposed to ethnography (where the researcher is an insider but does not change the environment), in AR the researcher is an insider and changes the environment.

- It is a qualitative, in situ and unstructured research method. It uses induction, but its generalization is difficult because it is context-specific.

There are three main types of AR:

Technical AR The researcher intervenes based on his technical expertise, sharing his knowledge with the users to solve a specific problem.

Practical-deliverative AR The researcher tries to understand the users' needs and requirements, and then tries to solve the problem based on that understanding (not necessarily based on his previous technical expertise).

Critical-emancipatory AR The researcher tries to understand the users' needs and requirements, and then tries to empower the users to solve the problem by themselves, without necessarily providing a solution directly.

AR involves a cyclical process of planning, acting and examining the results, with feedback loops that allow for continuous improvement and adaptation of the interventions. The three main steps of AR are:

1. Planning: The researcher identifies a problem and develops a plan to address it. *"Unfreezing"* the current state of the problem is crucial to allow for change, as it involves breaking down existing habits and behaviors that may be resistant to change.

2. Acting: The researcher implements the plan and takes action to address the problem. This may involve introducing new technologies, processes, or interventions to improve the situation.

3. Examining the results: The researcher evaluates the outcomes of the actions taken and reflects on the effectiveness of the interventions. *"Refreezing"* the new state of the problem is important to ensure that the changes are sustained over time, as it involves reinforcing new habits and behaviors that have been established.

1.3. Research through Design

RtD is a research method that is based on the idea of using design as a means of inquiry and exploration. It involves creating prototypes and artifacts not to solve a specific problem, but to gain insights and understanding about the design space and the users' needs and requirements.

- It is a qualitative, in situ and usually unstructured research method.

It is a cyclical process that involves four main steps:

1. Selecting a problem: The researcher identifies a problem or area of interest to explore through design.

2. Designing a solution: The researcher creates a prototype or artifact that addresses the selected problem or area of interest.

3. Evaluating the solution: The researcher evaluates the prototype or artifact to gain insights into the design space and the users' needs and requirements.

4. Reflecting on the results: The researcher reflects on the insights gained from the evaluation and uses them to inform the next iteration of the design process.

It has several benefits, such as:

- It shows technical opportunities and challenges that engineers may not have thought of.

- Exposes the gaps between the theory and practice of design.

- Changes current practices, therefore opening up new design spaces.

- Reveals designer patterns and assumptions, which can be used to inform future design decisions.

1.4. Field Deployment

Field deployment is a research method that involves deploying a prototype or artifact in the users' natural environment to gain insights into its use and effectiveness. It allows researchers to observe how users interact with the prototype in real-world settings, providing valuable feedback for further design iterations. Testing it in the lab is not enough, as it may not capture the complexities and nuances of real-world use.

- It is a qualitative, in situ and usually structured research method.

There are 7 main steps in field deployment:

1. Finding a Setting: There are three different types of settings:
    - *Convenience*: Deployment within the researcher's own environment. Not representative, and participants may be biased.
    - *Semi-Controlled*: It involves knwon and unknown participants. It can be more representative, but it may still have some biases.
    - *In the Wild*: Deployment in the users' natural environment to almost unknown participants. It is the most representative, but needs more robust prototypes.
2. Defining Goals: The fundamental goal is increasing understanding of the prototype that generalizes beyond the specific deployment.
3. Recruiting Participants: They should be representative of the target user population, also taking into account the researcher's group.
4. Designing Data Collection Instruments: Both qualitative (e.g., interviews, observations) and quantitative (e.g., measured time to complete a task, error rates) data collection instruments should be designed to gather comprehensive insights into the users' interactions with the prototype.
5. Conducting the Deployment: Pilot tests should be conducted to identify and address any issues with the prototype or data collection instruments before the full deployment.
6. Ending the Deployment: When the deployment is over, the product is usually taken away from the users in order to reduce the cost of mantaining its servers. In addition, they are usually not allowed to be bought by the users, as usually researchers do not want to sell products but only to gain insights.

    Taking the prototype away from the users can be a difficult process, as it may have become an integral part of their daily lives. It is important to plan for this transition and provide support to users as they adjust to the change.
7. Analyzing the Data

1.5. Experimental Research

Experimental research is a research method whose aim is to know the cause-effect relationship between two or more variables. It involves manipulating one variable (called *independent variable*) and observing the effect on one or more other variables (called *dependent variables*), while controlling for other factors that may influence the results. In order to conduct a good experiment, a prediction (called *hypothesis*) should be made about the expected relationship between the variables.

- It is a quantitative, in laboratory and structured research method.

Two different factors should be taken into account when designing an experiment:

- Validity: It refers to the extent to which the results of an experiment can be generalized to other settings, populations, and times. There are several types of validity, as population validity, ecological validity, temporal validity, etc.
- Reliability: It refers to the consistency and stability of the results of an experiment. An experiment is considered reliable if it produces the same results when repeated under the same conditions.

As explained before, hypotheses are predictions about the expected relationship between variables, and they are essential for designing and conducting experiments. They should be *precise*, *meaningful* (based on theory), *testable*, and *falsifiable*. In order to test a hypothesis, the null hypothesis (which states that there is no relationship between the variables) should be rejected.

There are two different approaches to research in experimental research:

- *Between-Subject Design*: Different groups of participants are exposed to different conditions of the independent variable. This design allows for comparisons between groups, but it may require a larger sample size to achieve statistical power.
- *Within-Subject Design*: The same group of participants is exposed to all conditions of the independent variable. They are firstly exposed to one condition, and then to the other condition. This design allows for comparisons within the same group, but it may be affected by order effects (e.g., fatigue, learning).

  In order to mitigate order effects in within-subject designs, counterbalancing can be used, which involves varying the order of conditions across participants to ensure that any potential order effects are evenly distributed and do not bias the results. There are two main types of counterbalancing:

  - *Complete Counterbalancing*: All possible orders of conditions are used, which can be impractical for a large number of conditions, as $x!$ orders are needed (where x is the number of conditions).
  - *Latin Square Counterbalancing*: A more efficient method that uses a Latin square design to ensure that each condition appears in each position an equal number of times, reducing the number of orders needed to x (where x is the number of conditions).

1.6. Survey Research

Survey research is a research method that involves collecting data from a sample of individuals using standardized questionnaires or interviews.

- It is a quantitative, both in situ and in laboratory, and a structured research method.

Surveys are used to gather information about people's attitudes, beliefs, behaviors, and characteristics... but are not good when understanding the reasons behind those attitudes, beliefs, behaviors, and characteristics. Therefore, they are often used in combination with other research methods (e.g., ethnography, action research) to gain a more comprehensive understanding of the design space.

- Surveys are used to decide if the results of the research methods can be generalized to a larger population.

- The other research methods are used to understand the reasons or trends behind the attitudes, beliefs, behaviors, and characteristics of the users.

There are different stages in survey research:

1. Research Goals and Constructs: Firstly, the research goals should be defined, which will guide the entire survey research process. Then, the constructs that will be measured should be identified.

2. Population and Sampling: There are different scopes of people that can be surveyed. The population groups considered, from the most general to the most specific, are:

   - *Population*: The entire group of individuals that the researcher is interested in studying.
   - *Sampling Frame*: The individuals that the researcher has access to and can potentially include in the survey.
   - *Sample*: The individuals that are actually asked to participate in the survey. Not all individuals in the sampling frame may be included in the sample, as some may be excluded due to various reasons (e.g., ethical reasons, ineligibility). There are different sampling methods that can be used to select the sample:
     - *Probability Sampling*: Each individual in the sampling frame has the same probability of being selected for the sample. This method allows for generalization of the results to the population, but it may be more time-consuming and expensive.
     - *Non-Probability Sampling*: Individuals are selected based on non-random criteria, such as convenience or judgment. This method is less time-consuming and expensive, but it may introduce bias and limit the generalizability of the results.

- *Respondents*: The individuals that actually respond to the survey. The response rate (i.e., the percentage of respondents out of the total sample) is an important factor to consider, as a low response rate can lead to non-response bias and affect the generalizability of the results.

3. Questionnaire Design and Bias:

   There are two main types of questions that can be used in a survey:

   - *Close Ended Questions*: These questions provide respondents with a set of predefined response options to choose from. They are easier to analyze and easier for respondents to answer, but they may limit the depth and richness of the data collected. This should be the most used type of question in surveys. They can be further categorized into:
     - Single-choice questions.
     - Multiple-choice questions.
     - Likert scale questions, aka rating questions.
     - Ranking questions.
   - *Open Ended Questions*: These questions allow respondents to provide their own answers in their own words. They can provide richer and more detailed data, but they may be more difficult to analyze and may require more time for respondents to answer.

   When designing a questionnaire, it is important to be aware of potential biases that can affect the validity of the results. A bias is a systematic error that can lead to inaccurate or misleading results. Some common types of bias in survey research include:

   - *Satisficing Bias*: Respondents may use a low cognitive effort strategy to answer survey questions, such as selecting the first acceptable response option.
   - *Acquiescence Bias*: Respondents may have a tendency to agree with statements or questions, regardless of their true feelings or opinions. Questions with `agree/disagree` response options are particularly susceptible to this bias.
   - *Social Desirability Bias*: Respondents may provide answers that they believe are socially acceptable or desirable, rather than their true thoughts or feelings.
   - *Response Order Bias*: The order in which response options are presented can influence respondents' choices, with a tendency to select options that are presented first or last.
   - *Question Order Bias*: The order of questions can influence respondents' answers, as earlier questions may prime certain thoughts or feelings that affect responses to later questions.

   In addition, when designing a questionnaire, there are already existing validated questionnaires that can be used to measure certain constructs. Using these existing questionnaires can save time and effort, and can also improve the validity and reliability of the results, as these questionnaires have already been tested and validated in previous research.

4. Review and Survey Pretesting:

   There are two main types of pretesting that can be conducted to evaluate the questionnaire before it is administered to the full sample:

   - *Cognitive Pretesting*: This involves asking a small group of respondents to complete the questionnaire while thinking aloud, allowing the researcher to identify any issues with question wording, comprehension, or response options.
   - *Pilot Testing*: This involves administering the questionnaire to a small sample of respondents that are similar to the target population, allowing the researcher to identify any issues with the survey administration process, such as the length of time it takes to complete the survey or any technical issues with the survey platform.

5. Implementation and Launch:

   There are different well known platforms that can be used to implement and launch a survey, such as Qualtrics, SurveyMonkey, Google Forms, etc.

6. Data Analysis and Reporting:

   When analyzing survey data, it is important to consider the type of data collected (e.g., nominal, ordinal, interval, ratio) and to use appropriate statistical methods for analysis. Some additional metrics can be analyzed, such as:

   - *Data Logging*: Some survey platforms allow for data logging, which involves recording additional information about respondents' interactions with the survey, such as the time taken to complete the survey or the order of responses. This can provide valuable insights into respondents' behavior and can help identify potential issues with the survey design.
   - *Response Rates*: Analyzing response rates can help identify potential biases in the sample and can inform strategies for improving response rates in future surveys.

2. Human Robot Interaction

During this chapter, we will be talking about the interaction between humans and robots, and how to design interfaces that allow for effective communication and collaboration between the two. However, before we dive into it, it is important to understand what a *robot* is. A robot is a multi-purpose machine that can be programmed to perform a variety of tasks emulating living beings. They are therefore mobile and autonomous. The three esential aspects of a robot are:

- Sensing: The ability to perceive the environment through sensors, such as cameras, microphones, and touch sensors.
- Thinking: The ability to process information and make decisions based on the data received from the sensors.
- Acting: The ability to perform actions based on the decisions made by the robot, such as moving, manipulating objects, or communicating with humans.

Since robots are becoming more common in our daily lives, it is crucial to correctly understand the Human Robot Interaction (HRI), specially because of the *social* aspect of it. Even though robots may have a correct physical interaction because of their sensors and actuators, they may fail to interact with humans in a way that is socially acceptable or effective. HRI is a multidisciplinary field therefore focuses on developping robots that can interact with humans in a natural and intuitive way.

2.1. Classifications in HRI

When working in HRI, plenty of different classifications should be taken into account. However, the most important ones are:

- Robot classification depending on their interaction capabilities.
- Human classification depending on their role in the interaction.
- Interaction classification depending on the type of interaction between humans and robots.

Depending on their interaction capabilities, we can classify robots into different categories:

- Bounded Autonomy: Robots with limited autonomy that operate within predefined constraints.

- Teleoperation: Robots controlled remotely by a human operator.
- Supervised Autonomy: Robots that operate autonomously but under human supervision.
- Adaptive Autonomy: Robots that can adapt their behavior based on the environment and user needs.
- Virtual Symbiosis: Robots that collaborate with humans in a way that enhances both parties' capabilities. The robot is a co-worker.

Depending on the role of humans in the interaction, we can classify them into different categories:

- Supervisor: The human is responsible for overseeing the robot's actions and ensuring that it operates safely and effectively.
- Operator: The human is responsible for controlling the robot's actions, either directly or through a user interface.
- Mechanic: The human is responsible for maintaining and repairing the robot, ensuring that it remains in good working condition.
- Peer: The human is a collaborator with the robot, working together to achieve a common goal.
- Bystander: The human avoids interaction with the robot, either because they are not interested or because they are afraid of it.

Depending on the type of interaction between humans and robots, we can classify them into different categories:

- Coexistence: The human and the robot share the same environment at the same time, but they do not interact with each other.
- Cooperation: The human and the robot share the same environment at the same time, and they share a common goal, but they do not contact or interact with each other to achieve it.
- Collaboration: The human and the robot share the same environment at the same time, share a common goal, and they interact with each other to achieve it.

2.2. Development of a Robot

When developing a robot, there are several different aspects that should be taken into account. In order to design it, a modular approach should be taken, and the *three main components of the robot* are considered from the sense-think-act approach:

- Perception (Hardware and Software):

  Both self monitoring and environment perception are crucial for a robot to interact effectively with humans.

  - *Self monitoring*: The robot should be able to monitor its own state, such as its battery level, motor status, and sensor readings.
  - *Environment perception*: The robot should be able to perceive the environment through sensors, such as cameras, microphones, and touch sensors.

- Cognition (Software):

  The robot should have a cognitive architecture that allows it to process information and make decisions based on the data received from the sensors. It runs on operating systems, such as ROS (Robot Operating System), that provide a framework for developing and deploying robot software.

  Their skills are usually preprogrammed, but they can also learn from experience and adapt to new situations (machine learning). Most of their learning capabilities are offline, but active research is being done in online learning, which allows robots to learn and adapt in real-time.

- Action (Hardware and Software):

  In order to perform actions, the robot should have actuators, such as motors and grippers, that allow it to move and manipulate objects.

Once the three main components of the robot are defined, the design of the interaction can be done. Two different approaches are:

- Inside-out approach: The design starts by building the technology, and the interaction is designed around it. It is later tested by social scientists, who evaluate the interaction and provide feedback to the engineers.
- Outside-in approach: The design starts by understanding the social context and the needs of the users, and then the technology is developed to meet those needs.

When designing the robot, there are different aspects that should be considered regaring the interaction with humans. Some of the most important ones are:

- Morphology: The physical design of the robot, which can affect how humans perceive and interact with it. It should take into account several factors:
  - The form and their function should be consistent.
  - Android robots (those that physically resemble humans) should be designed with care, as they may be antropomorphic. In addition, the *uncanny valley* phenomenon should be taken into account (Figure 2.1), which describes the discomfort that humans may feel when interacting with robots that are very close to human appearance, but not quite there.
  - Antropomorphic robots (those that are attributed with human traits or emotions) should be designed with care, as they may create unrealistic expectations about their capabilities. There are several questionnaires that can be used to evaluate the antropomorphism of a robot, such as the Godspeed questionnaire.

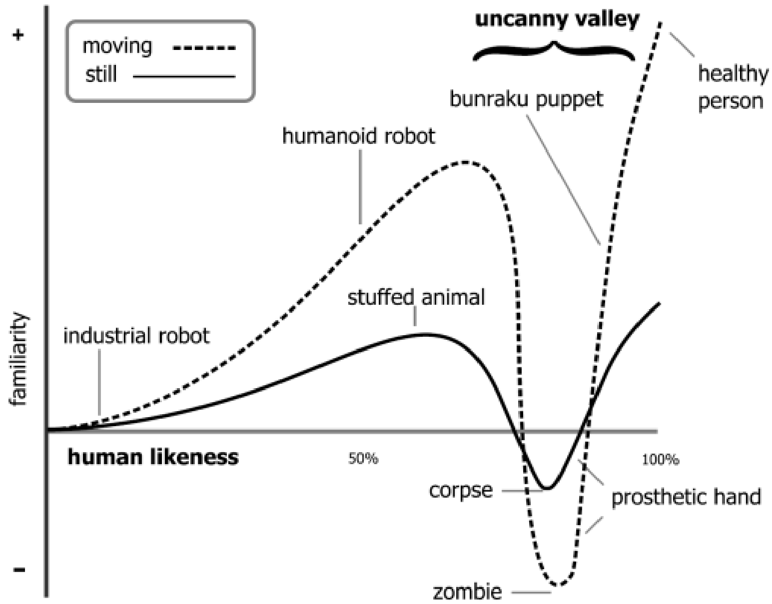

Figura 2.1: The uncanny valley phenomenon.

It should be noted that android are more tend to be antropomorphic, but these are not the same thing. For example, a robot that looks like a dog may be antropomorphic, but it is not an android. In addition, computers can also be antropomorphized when humans shout at them, but they are not robots.

- Affordances: The design of the robot should provide clear affordances that indicate how it can be interacted with. For example, a robot with a handle may indicate that it can be picked up, while a robot with a screen may indicate that it can be interacted with through touch.

- Underpromise and overdeliver: Expectations should be defined clearly, and the robot should be designed to exceed those expectations.

- Interaction expands functionality: On open-ended design should be taken, allowing for the interaction to expand the functionality of the robot beyond its initial design. For example, a robot designed for cleaning may also be able to provide companionship and social interaction.

- Human-centered design: The design of the robot should be centered around the needs and preferences of the users, using prototypes and user testing to iteratively improve the design.

- Culture: The design of the robot should take into account the cultural context in which it will be used, as different cultures may have different expectations and preferences regarding robots.

2.3. Interaction Modalities

When designing the interaction between humans and robots, different modalities can be used to facilitate communication and collaboration. In the following sections, we will talk about the most common interaction modalities used in HRI.

2.3.1. Spatial Interaction

Placing a robot in people space should consider people's preferences. There are plenty of aspects that should be taken into account, such as:

- Localization and Navigation: Robots should be able to localize themselves in the environment, so they should be able to build a map of the environment trough sensors, such as cameras and lidar, and use that map to navigate and avoid obstacles. They should also be able to localize their different body parts, such as their arms and grippers, to perform tasks that require manipulation.

- Social Positioning: Navigation in the environment is another challenge, as the robot should be able to navigate in a way that is socially acceptable and does not cause discomfort to humans. Proxemics defines the different zones of personal space that humans have, and robots should be designed to respect those zones.

- Initiating Interaction: Every interaction should be initiated. Although it may be trivial for humans, it is not for robots. For instance, frontal approach is more likely to initiate an interaction than a lateral approach.

- Intent Sharing: Robot motion trajectories should be designed to share the robot's intent with humans (even though they may not be the most efficient ones).

2.4. Nonverbal Interaction

Nonverbal communication is an essential aspect of human-robot interaction, as it can convey information and emotions that may not be easily expressed through verbal communication. The *Theory of Mind* is a crucial aspect of nonverbal interaction, as it is the ability to attribute mental states, such as beliefs, desires, and intentions, to oneself and others. Even though humans have a natural ability to understand the mental states of others (usually from a very young age), it is a challenge for robots to develop this ability.

Given this importance, robots should be designed to both express and understand nonverbal communication. Some of the most important types of nonverbal communication in HRI are:

- Gaze: Both humans' and robots' gaze can convey information about attention, interest, and intention. Robots should be designed to use their gaze and to interpret the gaze of humans to facilitate communication and collaboration.

- Gesture: As humans do, robots can use gestures to convey information and emotions. There are different types of gestures, such as:

- *Deitic gestures*: Gestures that refer to a specific object or location in the environment, such as pointing.
- *Iconic gestures*: Gestures that are used along with speech to convey information, such as using hand movements to illustrate a concept.
- *Symbolic gestures*: Gestures that have a specific meaning within a culture, such as a thumbs-up.
- *Beat gestures*: Gestures that are used related to the rhythm of speech.

- Touch: Even though touch is a powerful form of nonverbal communication, it is also the most difficult one to implement in robots as it means invading the personal space of humans. In some cases, they are however useful:
  - When the robot is zoomorphic, as it can be designed to be touched in a way that is similar to how humans interact with animals.
  - In nursing and caregiving robots, as touch can be used to provide comfort and support to patients.
- Posture and movement: The posture of the robot can also be used to convey information and emotions.

2.5. Human-Robot Teams

In some cases, humans and robots may need to work together as a team to achieve a common goal. Team members have the following characteristics:

- Share one or more common goals.
- Independently perform tasks that contribute to the common goal.
- Interact socially to coordinate their efforts and achieve the common goal.

Therefore, robots are also expected to be good team members. In order to know if a robot is felt as an actuall team member, two different concepts should be taken into account:

- Team Identity: Shared identity between a group of individuals.
- Team Identification: The degree to which an individual identifies (feels a sense of belonging) with a team.

Team Identification is crucial for effective teamwork, as it can lead to increased motivation, cooperation, and performance. Therefore, robots should be designed to foster team identification among human team members. *Trust* is also a crucial aspect of teamwork, so robots should be designed to build and maintain trust with human team members. There are different types of questionnaires that can be used to evaluate trust in the robot.

3. Human AI Interaction

AI systems are becoming increasingly prevalent in our daily lives, from virtual assistants to recommendation systems. As these systems become more sophisticated, it is crucial to understand how humans interact with them and how to design interfaces that facilitate effective communication and collaboration between humans and AI.

3.1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn. AI systems can perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation. Even though intelligence in humans can be *measured*, AI can't be measured. There are however various tests that have been proposed to evaluate if a machine can be considered intelligent. The importance of these tests is shown in the Chinese Room thought experiment.

- Chinese Room Experiment:

  The Chinese Room thought experiment involves a person who does not understand Chinese sitting in a room with a set of instructions for how to respond to Chinese characters. The person receives Chinese characters as input and uses the instructions to produce appropriate responses, which are then sent back out of the room.

  From the outside, it may appear that the person in the room understands Chinese, but in reality, they are simply following a set of rules without any comprehension of the language. This thought experiment raises questions about whether a machine can truly understand language or if it is simply following a set of instructions.

The most well-known test for evaluating machine intelligence is the Turing Test, proposed by Alan Turing:

- Turing Test:

  In the Turing Test, a human evaluator interacts with both a machine and a human without knowing which is which. If the evaluator cannot reliably distinguish between the machine and the human, then the machine is said to have passed the Turing Test and is considered to be intelligent.

  It only tests for the ability to mimic human behavior, not for true understanding or consciousness.

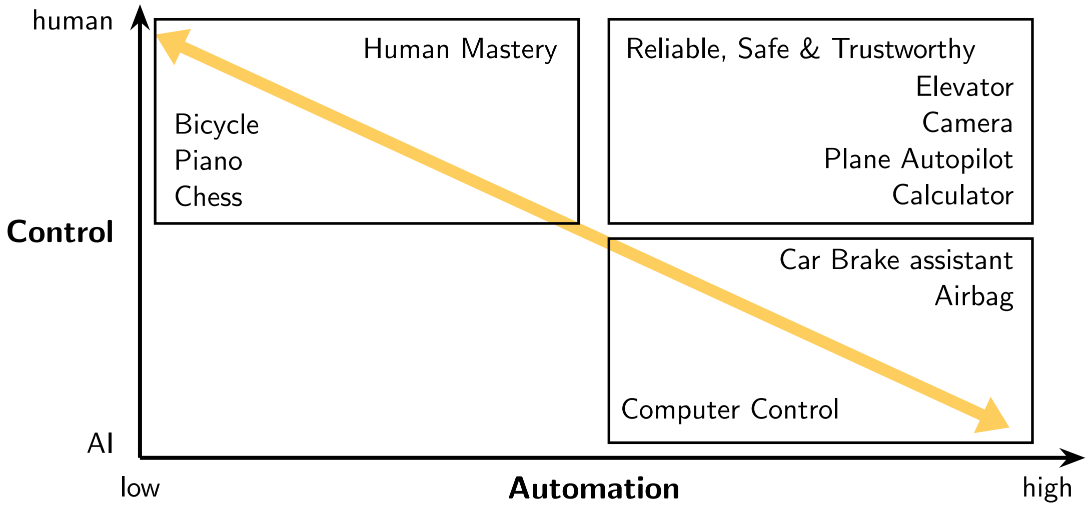

Figura 3.1: Control vs Automation in Human-AI Interaction

3.2. User's control

When designing AI systems, it is important to consider the user's control over the system. AI's goal is usually translating the raw data into a model that can be used to make predictions or decisions. However, there are different levels of control that users can have over this process:

- An algorithm that is fully automated does this conversion without any user input, so no human control is involved.
- Humans provide feedback to the training algorithm, which can be used to improve the model's performance. This is often referred to as "human-in-the-loop" learning.
- Users replace the training data or algorithms with their own knowledge and expertise, effectively taking all the control of the learning process, but having to do all the work.

Some researches argue that there is a *trade-off between the level of control and the automation of the system*. Fully automated systems may be more efficient and require less effort from users, but they may also be less transparent and less customizable. On the other hand, systems that allow for more user control may be more flexible and adaptable to individual needs, but they may also require more effort and expertise from users. However, some researchers argue that this trade-off is not necessarily inherent in the Human-AI interaction. This idea is shown in Figure 3.1, where the diagonal line represents the trade-off between control and automation, but there are also areas where both can be achieved simultaneously.

3.3. Explainability in AI Systems

Explainability refers to the ability of an AI system to provide clear and understandable explanations for its decisions and actions. There are several reasons why explainability is important in AI systems:

- Trust: Users are more likely to trust an AI system if they can understand how it works and why it makes certain decisions.
- Accountability: Explainable AI systems allow decisions to be audited and held accountable, which is especially important in high-stakes applications such as healthcare and finance.
- Ethical Considerations: Explainability can help ensure that AI systems are used ethically and responsibly, as it allows users to understand the potential consequences of the system's actions.
- Debugging and Improvement: Explainable AI systems can provide insights into how the system is functioning, which can be useful for debugging and improving the system over time.

There are different recipients of explanations in AI systems, including:

- Users affected by the model's decisions: These users may require explanations to understand how the model's decisions impact them and to ensure that they are treated fairly.
- Data scientists and developers: These individuals may require explanations to understand how the model is functioning and to identify potential issues or areas for improvement.
- Regulators and policymakers: These stakeholders may require explanations to ensure that AI systems are compliant with regulations and ethical standards.
- Domain experts or users of the system: These users may require explanations to understand how the AI system is making decisions in their specific domain and to ensure that it is providing accurate and relevant information.
- Managers and decision-makers: These individuals may require explanations to understand the implications of the AI system's decisions and to make informed decisions based on the system's outputs.

There is however an important trade-off between the accuracy of an AI system and its explainability. Highly accurate AI systems may be less transparent and harder for users to understand, which can lead to a lack of trust. On the other hand, more explainable AI systems may be less accurate but can foster greater trust among users. This is the goal of the field of Explainable AI (XAI), which aims to develop AI systems that are both accurate and explainable. Depeding on the application and the users, there are different types of explanations that may be more appropriate, such as:

- Analytical Explanations: These explanations provide a detailed analysis of the model's decision-making process, including the features and data that influenced the decision.
- Visual Explanations: These explanations use visualizations to help users understand the model's decision-making process, such as heatmaps or feature importance graphs.

- Rule-Based Explanations: These explanations provide a set of rules or guidelines that the model follows to make decisions, which can be easier for users to understand. It is presented as a simple `if-then` statement. It may be too simplistic for complex models, but it can be useful for certain applications where interpretability is more important than accuracy.

- Textual Explanations: These explanations provide a natural language description of the model's decision-making process, which can be more accessible for users who are not familiar with technical details. There are different types of textual explanations, such as:

  - Contrastive Explanations: These explanations focus on why a particular decision was made instead of another, highlighting the differences between the two outcomes.
  - Counterfactual Explanations: These explanations focus on what changes would need to be made to the input data to produce a different outcome, providing insights into the model's decision boundaries.

There are four steps that should be followed to design effective explanations for AI systems:

- Identify the target audience: Understanding who will be receiving the explanations is crucial for designing explanations that are appropriate and effective for that audience.

- Determine the purpose of the explanation: The purpose of the explanation will guide the content and format of the explanation, whether it is to build trust, provide accountability, or facilitate debugging and improvement.

- Choose the content of the explanation: The content of the explanation should be relevant and informative for the target audience, providing insights into the model's decision-making process without overwhelming users with technical details.

- Select the format of the explanation: The format of the explanation should be accessible and understandable for the target audience, whether it is through visualizations, textual descriptions, or rule-based explanations.

A last important consideration when designing explanations for AI systems is reliance on the AI's answers.

- Over-reliance: Users may become overly reliant on the AI system's outputs, leading to a lack of critical thinking and potential errors if the system provides incorrect information.

- Under-reliance: Conversely, users may under-rely on the AI system's outputs, leading to missed opportunities for improved decision-making and efficiency.

Appropriate explanations can help mitigate these issues by providing users with the necessary information to make informed decisions about when to trust the AI system and when to exercise caution.

4. Augmented Reality

Augmented Reality (AR) is a technology that superimposes computer-generated content onto the real world, enhancing the user's perception and interaction with their environment. There are two different types of environments:

- Real Environment: The physical world that we live in, which can be seen and interacted with directly.
- Virtual Environment: A computer-generated environment that may (or may not) resemble the real world, and is typically experienced through a device such as a headset or smartphone.

When the virtual and real environments are combined, we get what is known as Mixed Reality (MR). There are two different types of MR:

- Augmented Reality (AR): The virtual content is overlaid onto the real world, allowing users to see and interact with both the real and virtual elements simultaneously.
- Augmented Virtuality (AV): The real world is integrated into a virtual environment, allowing users to interact with real-world objects within a virtual space.

4.1. Devices for Augmented Reality

In order to experience Augmented Reality, users typically require specific devices that can capture the real world and display the augmented content. Some common devices for AR include Smartphones, Tablets, AR Glasses, and Head-Mounted Displays (HMDs). Each of these devices has its own advantages and limitations in terms of portability, field of view, and interaction capabilities. Some aspects to consider when choosing an AR device include:

Video vs Optical See-Through

- Video see-through (VST): devices use cameras to capture the real world, and then computer-generated content is superimposed onto the video feed.

  It can be implemented using a smartphone or tablet, where the device's camera captures the real world and the screen displays the augmented content. It also allows a full peripheral view of the real world, which may be needed in dangerous situations.

- Optical see-through (OST): devices have two layers: a transparent layer that allows users to see the real world directly, and a reflective layer that projects virtual content onto the transparent layer.

  OST devices typically require specialized hardware such as AR glasses or HMDs, which can be more expensive and less portable than VST devices. It can also provide a more immersive experience, but may suffer from latency and lower image quality compared to VST. Lastly, users may suffer from motion sickness or eye strain when using OST devices for extended periods of time.

HMD vs Mobile vs Projector

There are different specific devices for AR, each with its own advantages and disadvantages. Each one has a spatial relationship to the user and the real environment, which needs parameters to be known. There are two types of parameters:

- C: **C**onstant and **C**alibrated parameters that remain stable during usage.
- T: **T**racked parameters that change constantly and therefore need tracking.

Depending on the device, the parameters that need to be tracked may differ.

- HMDs used for OST: The distance to the real world needs to be tracked (T), while the distance to the eyes is constant (C). Depending if it uses eye-tracking or not, that mey be tracked (T) or constant (C).
- HMDs used for VST: The distance to the real world needs to be tracked (T), and the distance between the camera and the display is constant (C), as they are integrated. The distance to the eyes is either constant (C) or tracked (T), depending on whether the device uses eye-tracking or not.
- Mobile devices: The distance to the real world needs to be tracked (T), while the distance from the camera to the device is constant (C), as they are integrated. The distance to the eyes usually does not matter.
- Projectors: The eyes do not matter, as the content is projected onto a surface. The distance to the real world is either constant (C) or tracked (T), depending on whether the scene is static or dynamic.

Occularity in HMDs

- Monocular: The virtual content is displayed to only one eye, which can lead to a less immersive experience but is cheaper and less complex to implement.
- Bi-ocular: The virtual content is displayed to both eyes, but the same image is shown to each eye. This can provide a more immersive experience than monocular devices, but may still lack depth perception.

- Binocular: The virtual content is displayed to both eyes, with each eye receiving a slightly different image to create a sense of depth and immersion. This can provide the most immersive experience, but is also the most complex and expensive to implement.

Distance to the display

- Head Space: The display is located close to the user's head, such as in HMDs.
- Body Space: The display is located at a distance from the user's body, such as in mobile or handheld devices.
- World Space: The display is located in the environment, such as stationary displays or projectors.

Field of View (FOV)

The FOV is the extent of the observable world that can be seen at any given moment. A wider FOV can provide a more immersive experience, but may also require more processing power and higher-quality hardware to maintain image quality and reduce latency. Humans real FOV is usually larger than most AR devices can provide, which can lead to a less immersive experience and may cause users to feel disconnected from the augmented content.

4.2. Making sense of the real environment

In order to effectively augment the real environment, AR systems need to be able to understand and interpret the physical world. This involves several key components, as registration, tracking, and calibration.

4.2.1. Registration

Registration is the process of aligning the virtual content with the real world in a way that appears seamless and natural to the user. In order to do so, the real-world coordinate systems need to be mapped to the virtual coordinate systems. Several transformations are needed to achieve this mapping, including:

- Model Transformation: This transformation maps the virtual content from its own coordinate system to a common world coordinate system.

  If the real objects move and the virtual content needs to be registered to them, tracking is needed to update the model transformation in real time.

- View Transformation: This transformation maps the world coordinate system to the user's viewpoint, taking into account the user's position and orientation in the real world.

  If the camera is moving, tracking is needed to update the view transformation in real time.

- Projection Transformation: This transformation maps the 3D virtual content onto the 2D display of the AR device, taking into account the device's field of view and other display parameters.

  Calibration is needed to determine the parameters of the projection transformation.

Depending on where the objects are registered, there are different types of registration:

- Object Registration: The virtual content is registered to specific objects in the real world, such as a table or a building. If the objects are moving, tracking is needed to update the registration in real time.
- View Registration: The virtual content is registered to the user's viewpoint, so that it appears to be anchored in the real world regardless of the user's position or orientation. If the camera is moving, tracking is needed to update the registration in real time.
- World Registration: The virtual content is registered to a fixed location in the real world, such as a specific point on the ground or a landmark. Regardless of whether the user or the objects are moving, no tracking is needed to maintain the registration, as it is anchored to a fixed location in the real world.

In addition, there are two main types of registration techniques, depending on how well objects are registered:

- Geometric Registration: it focuses on just placing the virtual content in the correct position and orientation in the real world.
- Photometric Registration: it focuses on matching the lighting and shading of the virtual content to the real world, in order to create a more seamless and realistic integration.

  It is much more computationally expensive than geometric registration, as it requires real-time analysis of the lighting conditions in the real world and the ability to adjust the virtual content accordingly.

Lastly, *occlusion* should be considered in registration, as it can affect the realism of the augmented content. If virtual objects are not properly occluded by real objects, it can break the illusion of integration and make the augmented content appear less believable. Proper occlusion handling requires accurate depth sensing and tracking of both the real and virtual objects in the environment.

4.2.2. Tracking

Tracking is the process of continuously monitoring and measuring the real-world environment and the user's position and orientation. There are different types of tracking systems, including:

- Stationary Tracking: The tracking system is fixed in a specific location and tracks the user's position and orientation relative to that location.

- Mobile Tracking: The tracking system is integrated into the AR device and tracks the user's position and orientation relative to the real world.

- Optical Tracking: The tracking system uses cameras and computer vision algorithms to track the user's position and orientation based on visual markers or features in the real world.

There are two main approaches to tracking in AR:

- Model-Based Tracking: This approach uses a pre-defined model of the real world, such as a 3D map or a set of known landmarks, to track the user's position and orientation. The tracking system compares the real-world environment to the model and updates the user's position and orientation accordingly.

- Model-Free Tracking: This approach does not rely on a pre-defined model of the real world, but instead uses real-time sensor data to track the user's position and orientation.

Different specific tracking techniques include:

- Marker-Based Tracking: This technique uses visual markers, such as QR codes or fiducial markers, placed in the real world.

- Natural Feature Tracking: This technique uses computer vision algorithms to track natural features in the real world, such as edges, corners, or textures.

  An specific example of this technique is Simultaneous Localization and Mapping (SLAM), which combines:

  - Visual natural attribute tracking
  - Depth sensing using specialized sensors.

One specific object that usually needs to be correctly tracked is the user's hands and fingers, as they are often used for interaction with the virtual content.

4.2.3. Calibration

Calibration is the process of determining the parameters of the AR system, such as the position and orientation of the cameras, the field of view, and the display characteristics. Proper calibration is essential for accurate registration and tracking, as it ensures that the virtual content is correctly aligned with the real world and that the user's movements are accurately tracked.

4.3. Situated AR

An important aspect of AR is the concept of *situated AR*, opposite to conventional AR.

- Conventional Visualization: The virtual content is displayed without integration with the real world. This includes traditional displays, or even AR content that is simply overlaid on the real world.

  This includes context registered relatively to markers, as they are added to the real world but do not have a specific meaning in it.

- Situated Visualization: The virtual content is integrated into the real world in a way that is meaningful and relevant to the user's context.

Some situated AR techniques include:

- Annotations and Labels: Virtual content is used to provide additional information about real-world objects or locations, such as labels, descriptions, or instructions.

  They should not cover real-world objects nor overlap with other annotations.

- X-Ray Visualization:

  This technique allows users to see through real-world objects, providing a view of hidden or obscured content. Transparency should still be used to keep up occlusion and depth impression.

  Some specific examples of x-ray visualization include showing the internal components of a machine, or allowing users to see through walls to detect wires or pipes.

- Spatial Manipulation:

  This technique allows users to manipulate virtual content in the real world, such as moving, resizing, or rotating virtual objects. It can be used for tasks such as interior design, where users can visualize how furniture would look in a room before making a purchase.

- Information Filtering:

  This technique allows users to filter or hide certain real-world objects or information, in order to focus on specific aspects of the environment. There are two main types of information filtering:

  - Knowledge-Based Filtering: This technique uses the user's knowledge and preferences to filter information.

    For instance, combined with X-Ray Visualization, it can be used to show only specific types of information, such as showing only the water pipes in a building while hiding the electrical wires.

  - Spatial Filtering: Use geometric and geographic information to place information. For instance, it can be used to show information about nearby points of interest while hiding information about distant locations.

4.4. AR Interaction

Interaction in AR can be achieved through various input methods and output modalities.

4.4.1. Output Modelities

There are three main output modalities in AR:

- Head-Referenced Display: The virtual content is displayed in a way that is anchored to the user's FoV, such as in smart glasses.
- Hand-Referenced Display: The virtual content is displayed in the user's palm of hand(s), allowing for direct interaction with the augmented content.
- Agile Display: The virtual content is displayed everywhere and with changing perspectives of users, such as in mobile displays (carried around or worn) or stationary displays (adaptive projectors).

Avatars

Avatars are virtual representations of users in the AR environment, allowing for social interaction and communication with other users. Some different aspects that should be taken into account when designing avatars include:

- Full or half body representation.
- Full featured or simplified abstract representation.

  Even though a full featured avatar can provide a more immersive experience and are usually more trusted, they can also be more expensive and complex to implement, and may require more processing power and higher-quality hardware to maintain image quality and reduce latency.
- Life size or scaled representation.

  Miniature avatars are usually preferred over life-size ones, and a potential reason for this is that they can fit in the user's FoV more easily.

4.4.2. Input Modalities

There are several input modalities for AR interaction, including:

- Rigid Object Tracking: This technique tracks the position and orientation of your head (HMD) or different devices (mobile, projectors) to allow for interaction with the virtual content.
- Gesture Recognition: Using hand tracking, the system can recognize specific hand gestures to trigger interactions with the virtual content. This gesture recognition should be as intuitive and natural as possible.
- Pointing: A laser beam or a ray can be emitted from the user's hand or device to point at the AR scene, allowing for selection of objects and non-verbal communication.
- Drawing: Similar to pointing, but instead of just selecting objects, the user can draw in the air to create virtual content or annotate the real world.
- Touch: As in real interfaces, virtual buttons or controls can be placed in the AR scene, allowing users to interact with them through touch.

The Midas Touch Problem

An important issue in AR interaction is selecting objects in the AR scene.

- Selection via Gaze

  A first approach to selection was using the user's gaze, where the system tracked the user's eye movements to determine where they were looking in the AR scene.

  However, this lead to the *Midas*[1] *Touch Problem*, where every object that the user looked at would be selected, making it difficult to interact with the AR content without accidentally selecting things.

- Selection via Pointing or Gestures

  In order to solve the Midas Touch Problem, selection can be done through pointing or gestures, where the user explicitly indicates which object they want to interact with.

[1] King Midas was a figure in Greek mythology who was granted the ability to turn everything he touched into gold. Even though this power was initially seen as a blessing, it quickly became a curse, as Midas could not eat, drink, or interact with anything without turning it into gold.